CRYSTAL CHENG

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EDUCATION

University of California, Berkeley, Berkeley, CA — Expected Dec 2026

Bachelor of Arts in Computer Science & Applied Mathematics

- Coursework: Artificial Intelligence, Machine Learning, Data Structures, Statistical Modeling, Linear Algebra, Probability Theory
- Involvements: Society of Women Engineers (Treasurer), Web Development at Berkeley (Developer)

WORK EXPERIENCE

Hyve Solutions | Software Engineer Intern — June 2026 - Aug 2026

- Unblocked a **GenAI** app on Azure OpenAI hitting token limits by replacing full-catalog retrieval with a **RAG pipeline** (Hugging Face embeddings + Neo4j vector search), cutting per-request LLM candidates **~95%** via a top-150 cap with a keyword fallback.
- Shipped a full-stack RAG test-case search API using tiered keyword matching, RapidFuzz prefiltering, and LLM re-ranking behind an ID-allowlist guardrail that eliminated **hallucinated results**, adopted as Analyze's shared retrieval layer.
- Merged duplicated case-detail code from 2 navigation surfaces into a single context-aware renderer that switches context via a page-state flag, so the test-plan and test-case managers share one detail view with **zero logic duplication**.

TrackFly | Software Engineer Intern — Sep 2025 - Dec 2025

- Reduced cross-team integration mismatches on a **15-person cross-functional team** by designing REST API contracts and MySQL schemas that defined a single source of truth for feature data shared across microservices.
- Sped up delivery by shipping **15+ reusable React/TypeScript components** that standardized UI patterns over 10 weeks.

Barobo | Software Engineer Intern — May 2025 - Aug 2025

- Eliminated **unauthorized video sharing** across 100% of user traffic by engineering single-use AWS CloudFront signed URLs with per-user access control and configurable TTL expiration, so each link works for one authorized user.
- Scaled to **10,000+ monthly authorized video requests** with no added infrastructure overhead by tracking per-user access metadata in AWS DynamoDB, enabling reliable per-request authorization for every user's video playback request.
- Cut environment provisioning from **2 hours to 15 minutes** by rewriting deployment as reusable Terraform + AWS CLI scripts.

Opal Mentorship App | Machine Learning Intern — May 2024 - May 2025

- Raised mentor-mentee match **model** accuracy 12% across cohorts with 2 rounds of data-driven experiments on 6 scoring weights.
- Deployed a reproducible **ML evaluation pipeline** on AWS Lambda and DynamoDB scoring compatibility across 4 behavioral dimensions, versioning each weighting experiment so every model change stays auditable at cohort scale.

PROJECTS

RepoRank GitHub Search Engine | *Python, FastAPI, SQLAlchemy* — *link*

- Served BM25/BM25F search over 157k repos at **p95 23.8ms** uncached and sub-ms cached median latency (**94.3% hit rate**) on a from-scratch field-aware inverted index; benchmarked throughput to a ~100 QPS GIL-bound ceiling.
- Blocked ranking regressions before merge with an offline **evaluation** gate (nDCG/MRR/P@k with bootstrap confidence intervals over hand-labeled queries) wired into CI against a frozen 157k-repo index, failing builds on nDCG drops.
- Indexed **157k repos** by building a resumable, rate-limited GitHub crawler that recovers from interruptions and GitHub API rate limits, then rebuilds a fresh snapshot of the field-aware inverted index from every completed crawl run.

FreshCheck Fruit Freshness Classifier | *PyTorch, ONNX, React, Vercel* — *link*

- Caught **train/test leakage** from duplicate source images and re-split the full dataset on content and perceptual hashes into a leak-free held-out set, reporting honest 97.7% held-out accuracy with a 95% confidence interval.
- Ran **~14ms in-browser inference** with zero backend by training a 24K-parameter CNN in PyTorch and exporting it to ONNX (WebAssembly), keeping every user image on-device and removing all server hosting and inference costs.

MBTI Radar | *Hugging Face Transformers BART-MNLI, DeepFace, OpenCV, Python* — *link*

- Built a **multimodal classifier** on Hugging Face Spaces inferring all four MBTI axes via zero-shot BART-MNLI, late-fusing text, photo (DeepFace + OpenCV), and numeric signals with weights that redistribute to text when no photo is given.
- Corrected skewed predictions by diagnosing **identity-framing bias** in the zero-shot NLI hypothesis template, rewriting it around evidence-framing language, and adding a confidence-gap threshold so ambiguous axes abstain instead of guessing.
- Validated the text pipeline on a hand-written, unambiguous test set with **zero misclassifications** at >97% mean confidence.

SKILLS

- Languages: Python, C++/C, Java, JavaScript, TypeScript, Go, Bash, SQL, Ruby, HTML/CSS (Tailwind)
- ML & AI: PyTorch, vLLM, Triton, RAG, LLM Agents & Tool Calling, LLM-as-judge Eval, ONNX, TensorFlow, Scikit-learn
- Backend & Data: FastAPI, Node.js, REST APIs, Microservices, PostgreSQL, DynamoDB, AWS (S3, CloudFront, Lambda)
- Systems: Kubernetes, Docker, CI/CD (GitHub Actions, Jenkins), Terraform, Prometheus, Linux, Git, React, Next.js, pytest